Method of expressing concentration of solution

101. (c)Molarity $= \frac{w\times1000}{ml\text{ wt.}\times\text{Volume } ml.} = \frac{7.1\times1000}{142\times100} = 0.5M.$

102. (d)$M = \frac{4\times10}{40} = 1M.$

103. (d)Mole fraction $X = \frac{n}{n+N} = \frac{\frac{6}{60}}{\frac{6}{60}+\frac{180}{18}} = \frac{0.1}{10.1}.$

104. (c)$N = \frac{w\times1000}{Eq.wt.\times\text{Volume}} = \frac{10\times1000}{60\times100} = 1.66N.$

106. (b)$N = M \times$bosicity ; $N = 2\times2 = 4.$

108. (b)Concentration $= \frac{5\times10^6}{10^6} = 5\ ppm.$

110. (a) H_3PO_3 is a dibasic acid

N_1V_1 (acid) $= N_2V_2$ (base)

$0.1\times2\times20 = 0.1\times1\times V_2$

$\therefore V_2 = \frac{0.1\times2\times20}{0.1\times1} = 40ml$

111. (d)$H_3PO_4 \rightleftharpoons H^+ + H_2PO_4^-$

$H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-}$

$HPO_4^{2-} \rightleftharpoons H^+ + PO_4^{3-}$

Phosphoric acid does not give $1N$ strength.

112. (b)$C_6H_5COOH + NaOH \rightarrow C_6H_5COONa + H_2O$

$\frac{w}{40} = \frac{12.2}{122} = 4gms.$

113. (c) $^{(H_2SO_4)}N_1V_1 = N_2V_2$ (dilute acid)

$N_2 = (10 \times 36)/1000 = 0.36N.$

114. (c) $H_2O_2 \rightarrow H_2O + \frac{1}{2}O_2$

1 M H_2O_2solution $= 2N = 34gm/litre = 11.2$

So Normality$= \frac{2\times10}{11.2} = 1.75$

115. (b)Weight = molarity $\times$ $m.wt.\times$ v = $1 \times 132 \times 2 = 264gm.$

116. (a)Mole fraction $= \frac{n}{n+N} = \frac{\frac{w}{m}}{\frac{w}{m}+\frac{W}{M}} = \frac{\frac{1}{2}}{\frac{1}{2}+\frac{8}{32}} = 0.667.$

118. (c) 98% H_2SO_4 means 98g H_2SO_4 in 100g solution.

$\frac{100}{1.84}cc = 54.3cc;\ \ 98gH_2SO_4 = 1mol$

Hence molarity $= \frac{1}{54.3} \times 1000 = 18.4M$

120. (d) $3CaCl_2 + 2Na_3PO_4 \rightarrow Ca_3(PO_4)_2 + 6NaCl$

$\therefore$Mole of Na_3PO_4 = 3 mole of $CaCl_2$ = 1$mole$ $Ca_3(PO_4)_2$

$\therefore$0.2 mole of Na_3PO_4 = 0.3 mole of $CaCl_2$ = 0.1 mole of $Ca_3(PO_4)_2$.